

OPEN TO 2026 RESEARCH INTERNSHIPS

GRADUATING 2027

SEEKING 2027 FULL-TIME RESEARCH ROLES

500+	13	3	2
Google Scholar citations	first/co-first-author publications	panoramic · 3D · remote sensing	SemanticUrban · IndoorMS benchmarks

Research profile. Geometry-aware visual perception across panoramic imagery, 3D point clouds, and remote-sensing data, with an emphasis on sphere-native representation learning, robust dense prediction, multimodal fusion, and real-world benchmark construction for embodied and environmental intelligence.

RESEARCH STRENGTHS

Panoramic & embodied perception
Sphere-native features, rotation-robust dense prediction, and panoramic video/pedestrian tracking for robotic environment understanding.

3D scene intelligence
Point-cloud augmentation, multimodal 3D learning, and fine-grained urban scene understanding.

Earth observation
Semantic segmentation and sensor fusion for aerial, satellite, and multispectral imagery.

Collaborative directions
Landslide susceptibility mapping (several manuscripts under review) and quantum deep learning (ongoing 128-qubit simulation algorithm research).

EDUCATION

Ph.D. in Computer Science Expected 2027
University of Liverpool & Xi'an Jiaotong-Liverpool University
Advisor: Dr. Lei Fan

MRes in Computer Science 2023
University of Liverpool, United Kingdom

ACADEMIC SERVICE

Journal reviewer: TNNLS, TCSVT, T-ITS, TNSE, ISPRS JPRS, PR, NN, ESWA, EAAI, Neurocomputing.

Conference reviewer: ICLR, AAAI, IJCAI, NeurIPS, WACV.

SELECTED RESEARCH & PUBLICATIONS

ONGOING	Panoramic Pedestrian Tracking: Synthetic and Real-World Benchmarks with a New Baseline Dataset construction and baseline development for panoramic video understanding.
IJCV UNDER REVIEW	Panoramic Scene Understanding: A Survey from Distortion-Aware Engineering to Sphere-Native Modeling Qinfeng Zhu, Lei Fan. arXiv, 2026.
IJCV UNDER REVIEW	SO3UFormer: Learning Intrinsic Spherical Features for Rotation-Robust Panoramic Dense Prediction Qinfeng Zhu, Yunxi Jiang, Lei Fan. arXiv, 2026.
ESWA 2026 CO-FIRST	Towards Accurate Urban Scene Understanding Using Point Clouds: The SemanticUrban Dataset Yuan Fang*, Qinfeng Zhu*, Yuanzhi Cai, Lei Fan. *Equal contribution.
PR 2024	Advancements in Point Cloud Data Augmentation for Deep Learning: A Survey Qinfeng Zhu, Lei Fan, Ningxin Weng.
NN 2026	ClassWise-CRF: Category-Specific Fusion for Enhanced Semantic Segmentation of Remote Sensing Imagery Qinfeng Zhu, Yunxi Jiang, Lei Fan.
IEEE JSTARS 2024	Rethinking Scanning Strategies with Vision Mamba in Semantic Segmentation of Remote Sensing Imagery Qinfeng Zhu, Yuan Fang, Yuanzhi Cai, Cheng Chen, Lei Fan.
IEEE SENS. J. 2025	IndoorMS: A Multispectral Dataset for Semantic Segmentation in Indoor Scene Understanding Qinfeng Zhu, Jingjing Xiao, Lei Fan.

OTHER PUBLICATIONS

A Generalised Pre-Training Strategy for Deep Learning Networks in Semantic Segmentation of Remotely Sensed Images CGI / LNCS, 2026.	Samba: Semantic Segmentation of Remotely Sensed Images with State Space Model Heliyon, 2024.
From Landslide Conditioning Factors to Satellite Embeddings: Evaluating the Utilisation of Google AlphaEarth for Landslide Susceptibility Mapping Using Deep Learning Under review, arXiv, 2026.	Evaluating the Impact of Point Cloud Colorization on Semantic Segmentation Accuracy IEEE ICVISP, 2024.
SwinMamba: A Hybrid Local-Global Mamba Framework for Enhancing Semantic Segmentation of Remotely Sensed Images Digital Signal Processing, 2026.	Seg-LSTM: Performance of xLSTM for Semantic Segmentation of Remotely Sensed Images SSIP, 2024.
Enhancing Environmental Monitoring Through Multispectral Imaging: The WasteMS Dataset for Semantic Segmentation of Lakeside Waste MMM, 2025.	Algorithm-Driven Extraction of Point Cloud Data Representing Bottom Flanges of Beams in a Complex Steel Frame Structure for Deformation Measurement Buildings, 2024.
MSCrackMamba: Leveraging Vision Mamba for Crack Detection in Fused Multispectral Imagery IEEE BDAI, 2025.	Complete publication record, projects, code, and datasets: zhuqinfeng1999.github.io